

Genesis_Medicine: an open-source autonomous in silico drug-discovery pipeline for Korean traditional medicine — 7-tool active core, 40+ adapter scaffold catalog, cost-aware multi-fidelity Bayesian-optimization scheduler with runtime-gated cascading tiers, and OpenFold3 / AQAffinity calibration integration

HanCheongWoo ^{1,2,3}

ORCID: [0009-0004-4805-8815](https://orcid.org/0009-0004-4805-8815)

¹ Genesis_Medicine Lab, Seoul, Republic of Korea ² HAN PREDICT, Inc.; <https://hanpredict.com> ³ Recover Korean Medicine Clinic; <https://recover-clinic.kr>

Code: https://github.com/crazat/genesis_medicine · Correspondence: admin@hanpredict.com

Manuscript type: Resource / perspective paper with system architecture, design philosophy, and embedded multi-fidelity Bayesian-optimization scheduler (cost-aware acquisition + scientific gates + runtime gates + queue-state persistence) cross-linked to the companion scheduler paper; **Target preprint:** bioRxiv (Bioinformatics) — migrated from ChemRxiv per scope-rejection letter 2026-04-30; **License:** CC-BY 4.0 **Status:** v0.4 (2026-05-03) — bioRxiv resubmission. ChemRxiv rejected for scope; v0.3 audit-corrected pipeline description preserved; expanded with cost-aware multi-fidelity BO scheduler (752-pair queue snapshot, 8 protein targets), OpenFold3 + AQAffinity calibration tier integration, Round 8 final 25-adapter inventory promoted from appendix to main results, and cross-link to the companion scheduler paper (preprint #18). System / pipeline description; underlying experimental validation reported elsewhere

Abstract

Modern AI-driven drug discovery is built on a heterogeneous stack of open-source tools. Specialized application domains require tool selection, integration, and adaptation. We describe **Genesis_Medicine**, an open-source pipeline for **AI-driven Korean traditional medicine (한약) drug discovery**. We honestly distinguish: (i) a **7-tool active core** that produces all real pipeline outputs in this work (Boltz-2 cofold, REINVENT 4 generative, ADMET-AI v2 property prediction, OpenMM 8 + openmmtools alchemical sampling, RDKit chemistry, ChEMBL read-only, Open Targets v4 GraphQL); and (ii) a **40+ adapter scaffold catalog** organized into 12 functional tiers, written as Python modules and exposed through a natural-language agent interface but **not yet actively used in primary pipeline runs** — including AlphaFold-3-class cofold ensemble (Chai-1, Protenix-v2, OpenFold-3), generative SOTA (PocketXMol, FlowMol3, DiffSBDD, TurboHopp), ML potentials (MACE-OFF24, AIMNet2), pose validation (PoseBusters), and specialized

adapters (PROTAC designer, chronotherapy, TxGNN repurposing). The active core supports the complete workflow from natural-product curation through corrected ABFE; the adapter scaffold provides ready integration points for prioritized future expansion. We discuss the design philosophy — 3-pillar institutional integration (HAN PREDICT, Inc. AI healthcare platform; Recover Korean Medicine Clinic), commitment to honest in silico reporting, and explicit method limitations including MMP-1 zinc handling and Boltz-2 binary-classifier vs IC₅₀ distinction. The pipeline is open-source under Apache-2.0 at https://github.com/crazat/genesis_medicine.

Keywords: open-source pipeline, Korean traditional medicine, AI drug discovery, REINVENT4, Boltz-2, ADMET-AI, ABFE, OpenMM, natural products, integrated stack.

Plain-language summary

Modern AI-driven drug discovery uses many specialized software tools, mostly free and open-source. Combining these tools into a working pipeline for a specific application — in our case, AI-driven Korean traditional herbal medicine drug discovery — requires careful selection and integration. We describe an open-source pipeline (Genesis_Medicine) that brings together about 50 tools to support the complete workflow from herb-derived compound to in silico drug-candidate report. The code is freely available for academic and commercial use. **This paper is a description of the pipeline; no clinical or laboratory measurements are reported here.**

1. Introduction

1.1 The fragmented AI drug-discovery stack

Modern AI drug discovery requires composition of multiple specialized tools, each excellent at one task and not designed for end-to-end integration. The recent open-source landscape is rich: Boltz-2, AlphaFold-3, Protenix for structure prediction; REINVENT4, SATURN, f-RAG for generative chemistry; ADMET-AI, Chemprop 2 for properties; OpenMM 8 + openmmtools for MD and free energy; PoseBusters, GNINA for pose validation; Open Targets and DGIdb for disease-target context. Each tool has its own input format, environment requirements, and output structure. End-to-end use requires careful glue-code, validation, and domain-specific configuration.

1.2 The Korean traditional-medicine domain

The Korean traditional-medicine (한의학) domain has specific requirements that are not fully addressed by general-purpose pipelines:

- **Natural-product chemistry** (saponins, terpenoids, anthraquinones, polyphenols, alkaloids) — atypical scaffolds for which Caucasian-cohort-trained models may underperform.
- **Korean Pharmacopoeia (KP) and Korean Herbal Pharmacopoeia (KHP)** — specific listings, quality standards, traditional indications.

- **복합 처방 (multi-component formulations)** — Korean medicine prescribes combinations, not single compounds; pipeline must support combination scoring.
- **헌법 / 체질 (constitutional medicine)** — patient stratification by traditional constitutional theory.
- **Regulatory: MFDS, 제56조, 한약** — Korean-specific compliance.

1.3 Genesis_Medicine: design goals

We assembled the Genesis_Medicine pipeline with the following goals:

1. **End-to-end integration** of generative → property → structure → MD → ABFE → manuscript.
2. **Korean traditional-medicine domain orientation:** KP / KHP herb-compound mapping, **한약** network analysis, Korean PGx panel.
3. **Honest in silico reporting:** TRIPOD-AI compliance, explicit limitations, retraction of incorrect prior results when discovered.
4. **Open-source under permissive license:** Apache-2.0; commercial-use compatible; avoids GPL viral.
5. **Natural-language interface:** a single LLM-agent that accepts natural-language requests and dispatches to the appropriate pipeline tier.
6. **3-pillar institutional integration:** with HAN PREDICT (AI healthcare platform) and Recover Korean Medicine Clinic; clinical feedback loop closed.

2. Pipeline architecture: active core + adapter catalog

2.0 Active modules (~25, with real pipeline outputs)

A v0.3 internal audit (2026-04-26) revised our earlier framing. The active set is broader than initially reported:

Core compute: Boltz-2 v0.6.1 (MIT) — cofold + affinity (240+ cofolds across rounds + 4 disease screens); **REINVENT 4** v4.4 (Apache-2.0) — mol2mol scaffold-hopping; **ADMET-AI** v2.0.1 (MIT) — 6 endpoints; **OpenMM 8.5.1 + openmmtools 0.26** (MIT) — 10 ns MD + 16-window alchemical replica exchange ABFE; **RDKit** (BSD-3); **PoseBusters** 0.6.5 (queued for cofold pose validation); **Chai-1r** (Apache-2.0, 2025-Q4 fully open) — installed for 2-way ensemble; **GFN2-xTB** via xtb-python 22.1 — semi-empirical QM single-point (EMB-3 -50.09 Hartree, Embelin -65.90 Hartree computed in this work).

Database integration: ChEMBL v34 (CC-BY-SA), Open Targets v4 GraphQL (CC0; 11 disease IDs + 15 target IDs production-cached pipeline).

Workflow + reproducibility: **Prefect 3** DAG framework (11-stage pipeline orchestration); **Docker** + docker-compose containerization; **pyproject.toml** with ruff + mypy strict-mode; pinned dependencies; **GitHub Actions CI/CD** (lint + test + auto-PDF-build for preprints); 16 test files / 95 unit tests with GPU/slow/network markers.

Domain-specific active modules: residence_time.py (Eyring + Jarzynski + steered MD); spatial_atlas.py (Visium CytAssist + Xenium IPF atlas); scfoundation_adapter.py

(with linear baseline check); hira_qaly.py (Korean health economics); stratum_corneum.py (ceramide/FFA/cholesterol bilayer builder); benchmark_baseline.py (ROC-AUC + Mann-Whitney U); logKp head (FDA 2326 LGBM, ≥ -2 topical gate); chronotherapy/jaoryuju.py (12-meridian + circadian); donguibogam_miner.py (Korean classical text mining); fpocket cryptic_scan (TGFB1/MMP1/CTGF allosteric scan).

Recently activated (2026-04-26): BioEmu 1.3.1 (Microsoft 2026 protein equilibrium ensembles); deeptime 0.4.5 + PyEMMA 2.5.12 (MSM-based binding kinetics); xtb-python 22.1 (GFN2-xTB QM correction); 5 missing-target MSAs (MITF, SRD5A1, SREBP1, FBN1, mTOR); production Open Targets agent dispatch.

Outputs: all .csv and .json files in pilot/ are products of this active module set.

2.1 Adapter scaffold catalog (40+ modules)

Tier 0-11 below describe Python modules registered in agents/all_tiers_tools.py exposing 70+ natural-language tool dispatches. **These are integration scaffolds**, mostly subprocess/Python wrappers over external open-source tools that have not yet been run in the present pipeline. They are documented here for reproducibility of the integration architecture, not as active capability claims.

Tier 0 — Foundation (~7 modules)

Open-source SOTA stack: Boltz-2 cofold, REINVENT 4 generative, ADMET-AI predictions, RDKit chemistry, OpenMM 8 MD, openmmtools alchemical sampling, MACE-OFF24 ML potential. Standard conda / pip / uv installation under Korean GPU stack (CUDA 12.8, RTX 5090 Blackwell).

Tier 1 — Acceleration / Quality (~6 modules)

Protenix-v2, OpenFold-3 ensemble (Boltz-2 consensus); AlphaFlow-Lit; CarsiDock-Cov for covalent-warhead screening; Boltz-ABFE direct path; AIMNet2 ML potential; DeepRetro retrosynthesis.

Tier 2 — Skin-specific Adapters (~6 modules)

PanDerm (skin lesion vision-language; integrated as skin-classification adapter); ChemCrow LLM agent wrapper; stratum-corneum MD setup; Bayesian SAR optimization; ESM3 protein function; scFoundation single-cell.

Tier 3 — Multi-target / Cryptic-site (~5 modules)

Four added skin-fibrosis targets (LPA1 / $\alpha\text{v}\beta 6$ / YAP1 / TAZ); DGAT herb-DDI module; QM/MM ABFE book-ending correction; spatial-transcriptomics adapter; mechanotransduction multi-tier analysis.

Tier 4 — Kinetics / SAR loops (~5 modules)

k_{off} residence-time MD setup; Chou-Talalay synergy + topical PBPK; self-improvement agent loop; Enamine REAL Tanimoto matching; Lunit / VUNO collaboration analysis.

Tier 5 — Causal / Stratified evidence (~5 modules)

Causal inference (DoWhy / EconML); papillary-reticular fibroblast stratification; skin microbiome AI; KEDD foundation-model design; federated-learning Korean-medicine-network architecture.

Tier 6 — Korean PGx + skin niche (~5 modules)

Korean PGx panel; mitochondrial PGC-1 α / TFAM; CellChat fibroblast-niche LR pairs; smartphone scar assessment; OliX siRNA collaboration.

Tier 7 — Partnership scaffolds (~5 modules)

ExoCoBio collaboration scaffold; EZH2 epigenetic adapter; NLRP3 inflammasome; Pro-Hyp triple combination; PBM LED Recover protocol.

Tier 8 — Reporting + Health Economics (~5 modules)

TRIPOD-AI checklist agent tool; SHAP / Grad-CAM explainability; ☐☐☐☐ herb miner; D+Q senolytic hair-follicle; HIRA QALY framework.

Tier 9 — Diagnostic + Specialized (~5 modules)

facial_dx integration bridge; PROTAC EMB-3 TGFB1 degrader designer; chronotherapy ☐☐☐☐; OCT bridge to facial_dx; engineered C. acnes probiotic; ESG / green-chemistry analyzer.

Tier 10 — Safety + MLOps + Knowledge (~5 modules)

Hallucination guard (jailbreak + medical fact citation); MLOps model registry (FDA PCCP); knowledge-graph completion; PA-EWC continual learning; multi-omics fusion.

Tier 11 — Frontier / Strategic (~6 modules)

NaFM natural-product foundation model; Bayesian adaptive trial design + n-of-1 + MFDS DTx; KIPRIS + USPTO patent prior-art; SkinAge clock; Chai-1 ensemble; JUMP-CP morphology; MFDS K-bio M&A landscape.

Total modules (Tier 0 through Tier 11): approximately 50, with substantial cross-tier dependencies. The natural-language agent interface (`src/genesis_medicine/agents/`) exposes all of these as 70 callable tools.

3. Design philosophy

3.1 Honest in silico reporting

A central commitment is **honesty about what is and is not validated**. Specific mechanisms:

- All preprints carry a **"in silico, wet-lab pending"** flag in the abstract.

- **TRIPOD-AI checklist** is run on each preprint as a self-audit.
- **Earlier incorrect results are explicitly retracted** when discovered (e.g., the earlier ABFE values on EMB-3 / Embelin under a complex-leg-only protocol [2]).
- **MMP-1 zinc handling caveat** is consistently flagged across all preprints involving MMP-1 [2,3].
- **Boltz-2 binary-classifier vs IC₅₀** distinction is consistently made.
- **narrative correction** (chemistry-based clarification of Embelia ribes / distinction from / [4].

3.2 Apache-2.0 open-source

The pipeline code is open-source under Apache-2.0, which is permissive (commercial-use compatible, no GPL viral, no requirement to open-source derivative works). This contrasts with some traditional-medicine database licenses (HERB CC-BY-NC, TCMSP non-commercial) that we explicitly exclude from commercial-build profiles. The license model supports both academic publication and Recover Korean Medicine Clinic's commercial use.

3.3 3-pillar institutional integration

The pipeline is operated across three affiliated entities: - **HAN PREDICT, Inc.** (hanpredict.com) — AI healthcare platform; provides Smart Charts AI EHR + 3D facial diagnostics infrastructure. - **Genesis_Medicine Lab** — R&D division; develops and maintains the pipeline. - **Recover Korean Medicine Clinic** — clinical practice (2026-08-15 opening); provides clinical-context feedback loop.

This 3-entity integration is described in detail in our companion workflow preprint [5].

3.4 Korean-specific orientation

The pipeline is oriented to the Korean clinical, regulatory, and traditional-medicine context: - **Regulatory**: PIPA + Korean AI Basic Act + + MFDS pathways. - **Pharmacopoeial**: KP (Korean Pharmacopoeia) + KHP (Korean Herbal Pharmacopoeia) + Donguibogam digitization. - **Linguistic**: Korean clinical-context comments and natural-language tool descriptions. - **PGx**: Korean-population variant frequencies in pharmacogenomic panel.

This orientation does not preclude international application — the pipeline modules are individually general-purpose — but it does motivate specific configurations and weight choices.

4. Scaling and reproducibility

4.1 Hardware

The pipeline runs on a single-GPU configuration (1 × NVIDIA RTX 5090, 32 GB, CUDA 12.8). For reference, a typical end-to-end EMB-3-style scaffold-hopping case study

(REINVENT4 100 samples + ADMET filter + Boltz-2 cofold 30 jobs + 10 ns MD + corrected ABFE 64 ns) runs in approximately 24-30 hours on this hardware.

4.2 Reproducibility

- **Conda environments** locked: genesis-md (MD-specific) and main .venv (Python 3.11).
- **Configuration as code**: conf/*.yaml covers all major pipeline parameters.
- **Random seeds** documented per-run.
- **Result archival**: per-run JSON outputs include configuration, version metadata, wall-clock, and key results.

4.3 License compatibility for commercial use

Two build profiles: - **Research** profile: includes HERB 2.0, TCMSP, BATMAN-TCM (non-commercial) for network-pharmacology research. - **Commercial** profile: excludes the above; relies only on permissively-licensed COCONUT 2.0 (CC0), NPASS 3.0 (CC-BY), NPatlas (CC-BY), Dr. Duke's database (USDA public domain). Boltz-2 (MIT), REINVENT 4 (Apache-2.0), ADMET-AI (MIT), OpenMM 8 (MIT), openmmtools (MIT), RDKit (BSD-3) are all commercial-use-compatible.

This dual-profile approach permits academic and commercial use with appropriate licensing hygiene.

5. Limitations and forward path

1. **No clinical efficacy data** in any of the application studies; all are in silico hypothesis-generation.
2. **MMP-1 zinc handling** is a known limitation; ZAFF integration is planned.
3. **NaFM weights** (natural-product foundation model) are not yet integrated; placeholder fallback to Morgan fingerprint embeddings.
4. **Chai-1 integration** is interface-only; not benchmarked against Boltz-2 + Protenix consensus.
5. **JUMP-CP morphology** integration is offline-mock; AWS S3 data download (~50 GB) is the next forward step.
6. **Patent prior-art search** is offline-heuristic; KIPRIS + USPTO API integration requires API keys.

The Genesis_Medicine pipeline is a living open-source project. Contributions, issue reports, and forks are welcome under Apache-2.0.

6. Conclusions

Genesis_Medicine is an open-source, end-to-end AI pipeline for Korean traditional medicine drug discovery, integrating approximately 50+ tools across 12 functional tiers under a unified natural-language agent interface. The pipeline supports REINVENT4 generative scaffold-hopping, ADMET filtering, Boltz-2 cofold, MD validation, and

corrected ABFE estimation, with explicit Korean traditional-medicine domain orientation, honest in silico reporting, and 3-pillar institutional integration with HAN PREDICT, Inc. (AI healthcare platform) and Recover Korean Medicine Clinic. The pipeline is available under Apache-2.0 at https://github.com/crazat/genesis_medicine.

We invite collaboration and contribution from the AI-drug-discovery community.

Acknowledgments / Contributions / Competing interests / Data availability

Same standard text. Code: https://github.com/crazat/genesis_medicine.

Figures

Figure 1. Genesis_Medicine 12-tier pipeline architecture overview. Approximately 50+ modules organized into 12 functional tiers from Tier 0 (Foundation: Boltz-2, REINVENT4, ADMET-AI, OpenMM, openmmtools, MACE-OFF24, RDKit) through Tier 11 (Frontier: NaFM, Bayesian trial design, patent prior-art, SkinAge clock, Chai-1, JUMP-CP, MFDS K-bio landscape). Apache-2.0 open-source.

Genesis_Medicine 12-tier pipeline (50+ modules)
Apache-2.0 open-source · github.com/crazat/genesis_medicine

Tier 0	Foundation Boltz-2, REINVENT4, ADMET-AI, OpenMM, openmmtools, MACE-OFF24, RDKit
Tier 1	Acceleration / Quality Protenix, OpenFold3, AlphaFlow-Lit, CarsiDock-Cov, Boltz-ABFE, DeepRetro, AIMNet2
Tier 2	Skin-specific Adapters PanDerm, ChemCrow agent, Stratum corneum MD, Bayesian SAR, ESM3, scFoundation
Tier 3	Multi-target / Cryptic LPA1/αvβ6/YAP1/TAZ, DGAT-DDI, QM/MM ABFE, Spatial-T, mechanotransduction
Tier 4	Kinetics / SAR loops k _{off} MD, Chou-Talalay PBPK, self-improvement loop, Enamine REAL, Lunit/VUNO
Tier 5	Causal / Stratified DoWhy/EconML, papillary-reticular fibroblast, microbiome AI, KEDD, federated-learning
Tier 6	Korean PGx + niche Korean PGx panel, mitochondrial, CellChat, smartphone, OliX siRNA
Tier 7	Partnership scaffolds ExoCoBio, EZH2 epigenetic, NLRP3, Pro-Hyp, PBM LED
Tier 8	Reporting + HE TRIPOD-AI, SHAP/Grad-CAM, 동의보감 miner, D+Q senolytic, HIRA QALY
Tier 9	Diagnostic + Specialized facial_dx integration, PROTAC, chronotherapy, OCT, engineered probiotic, ESG
Tier 10	Safety + MLOps + KG hallucination guard, MLOps registry, KG completion, PA-EWC, multi-omics fusion
Tier 11	Frontier / Strategic NaFM, Bayesian trial + DTx, KIPRIS+USPTO patent, SkinAge, Chai-1, JUMP-CP, MFDS K-bio

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v0.1 draft, 2026-04-26 · ~3,200 words · CC-BY 4.0

Round 8 audit — Final 25-adapter inventory (2026-04-27)

After 4 audit rounds (Round 5 +9, Round 7 +5, Round 8 +11), the system reached **25 adapters across 14 modules**, with 14 in PRODUCTION_USE and 30+ real-data artifacts.

Module breakdown

Free-energy & ML potentials (md/, 6 adapters): ABFE corrected protocol, OpenMM ML refine, Boltz-ABFE scaffold, AToM-OpenMM scaffold, AIMNet2 scaffold, CarsiDock-Cov (covalent docker, FIRST-of-class).

Generative chemistry (generation/, 4 adapters): REINVENT 4 wrapper, FlowMol3, DecomDiff, MolDAIS BoTorch SAASBO.

Dermatology-specific (dermatology/, 4 adapters): PBK Dermal HT (NIH/NIEHS public-domain), SARA-ICE (OECD TG 497-III), Skin Fibroblast Atlas (Reynolds 2025), PanDerm (research only).

Repurposing (repurposing/, 3 adapters): TxGNN, CellAwareGNN (TxGNN successor), CMap L1000.

Conformational ensemble (ensemble/, 3 adapters): AlphaFlow, BioEmu, PocketMiner.

Foundation models (foundation/, 2 adapters): ChemBERTa-3, Tahoe-100M.

Causal inference (causal/, 1 adapter): TwoSampleMR with 7 literature-validated MR results.

Clinical workflow (clinical/, 4 adapters): IRB protocol generator, CRO quote, MedSAM-2 scaffold, IRB consent forms.

Drug-target kinetics (kinetics/, 2 adapters): τ RAMD literature-validated, SEEKR2 milestone.

Polypharmacology (polypharmacology/, 2 adapters): SwissTargetPrediction literature-validated, Dealbreaker safety panel (7 classes: hERG, CYP3A4, AR, GR, ER, PXR, AhR).

DDI (ddi/, 2 adapters): DDInter 2.0 + 17-pair curated $\square$ - $\square$.

Topical formulation (formulation/, 3 adapters): CPE-DB, HSP vehicle selection, KCID Korean cosmetic ingredient gating.

PK-PD (pkpd/, 2 adapters): htk (US EPA public domain), Hill 4-parameter dose-response.

Protein design (protein_design/, 1 adapter): LigandMPNN scaffold for $\square$ metal-aware design.

Honest categorical scoring

- Adapters with PRODUCTION_USE + literature-validated real data: 14 (56%)
- Adapters with SCAFFOLD_ONLY (graceful unavailable until external repo cloned): 11 (44%)
- Hydra YAML configs: 19/25 (76%)
- Unit tests: 29 covering Round 5+7+8 adapters (100%)

From "50 active tools" to honest 25

The earlier preprint v0.1 narrative claimed "50+ active tools." Round 5 audit (#175 honest reframe) corrected this to "25 ACTIVE modules + 25 adapters/wrappers". This v0.2+ count of 25 adapters is the precise audit-verified number, with 14 producing real data and 11 acting as graceful scaffolds for future external-tool integration.

Unique system contributions

1. **First DL covalent docker integration** (CarsiDock-Cov, Apache-2.0) for natural-product Michael acceptors.
 2. **First Korean cosmetic regulatory adapter** (KCID + KFDA + EU CosIng cross-walk) for in silico $\rightarrow$ product launch.
 3. **Single-pipeline integration** of Boltz-2 + ChEMBL $\rho=0.72$ calibration + Chai-1 ensemble + PoseBusters 149-pose validation + ABFE T4L PASS + EMB-3 cycle closed + Round 5+7+8 application data + 7 MR causal evidence + curated DDI + KCID regulatory.
 4. **Honest negative result:** EMB-3 $\times$ MMP-1 ABFE +0.55 quantitatively confirms "MMP-1 minus zinc" caveat $\rightarrow$ ZAFF release-blocking.
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Use of AI tools in writing (ICMJE 2024 disclosure)

The author used Claude (Anthropic, Opus 4.7) for drafting initial manuscript sections, generating tables, and editorial support during the writing of this preprint. The author personally:

- Designed the research protocol and experimental scope
- Performed all computational experiments and pipeline executions
- Verified every factual claim and quantitative result
- Validated all citations and external references
- Took full responsibility for the final content

AI tools were **not** used to generate experimental data, original hypotheses, or analytical results. All computational outputs (Boltz-2 co-folding, MD trajectories, ABFE estimations, ADMET predictions) were produced by named open-source software described in the Methods section, not by AI assistant tools.

This disclosure follows the International Committee of Medical Journal Editors (ICMJE) 2024 recommendations on artificial intelligence use in scholarly writing.